

Recommendations for Continuation of the DAP System

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I. Digital Twin

I.A. General Suggestions

1. If the Capstone team decides to make an independent simulation where the D.T predicts the actuator states and the only input into the D.T is the pilot log parameters, you must have a big group of people working on how to simulate the controller. Since we're not allowed to look at any of the internal decision making of the Seaglider due to ITAR, the corrective action of the vehicle would have to be redesigned from the ground up. The 2026 A&A Capstone began working on controller logic, but due to time constraints we decided to downscale.
2. Learn how to use seaglider.pub immediately. That means knowing the difference between the .log and .eng files and familiarizing yourself with variable definitions according to the Parameters Reference Manual, Seaglider Pilot's Guide and the files format document. The best way to learn about this aspect of the SSC is to attend a piloting session in the pilot room when a deployment is happening.

I.B. Dynamics

Before proceeding, it should be mentioned that the extent to which these issues and recommendations would affect the model is unknown; isolation of the effects of one issue would be difficult without an otherwise functional simulation. One should be proactive in leveraging their resources: TA's, mentors, professors, coworkers, and friends can and should provide feedback and guidance. This 6-DoF simulation task is beyond the scope of undergraduate classes, and assistance will be required.

Physical Parameters and Assumptions

There were a few simplifying assumptions made that may affect the model to differing degrees:

1. The center of buoyancy (CB) is assumed to be fixed at its position found in the SG 194 Trim Sheet. Inflation and deflation of the bladder will cause the CB to shift. To what extent this affects the model is unknown, but the next team may find it useful to investigate the changes this would produce.
2. The variable weight added for trimming the glider is also assumed to be constant, and at the value found in the SG 194 Trim Sheet, and it was incorporated into the static

mass instead of having its own variable. This mass may vary from mission to mission, and would need to be updated in this case.

3. The change in volume resulting from wing damage modes was not modeled. This would affect the overall volume of the glider, the mass, the location of the CB, and the location of the center of gravity (CG) of the static mass.

Most physical parameters were pulled from SG 194's October 2013 Trim and Balance Sheet. These parameters are likely out of date, and accessing a more recent trim and balance sheet is strongly recommended. The effects of this will propagate into inertia matrices, buoyancy, mass, CG location, and even the location of the CB. This will affect the entire coordinate reference frame.

It is recommended to find the distance that is set as neutral for the movable battery for each mission.

Numerical Integration

There may be issues with the choice of ODE solvers. ODE45 was used as the standard, but, on certain dives where the simulation broke down, ODE15s became necessary. The variable step size of these solvers was thought to compound issues. A test with 4th order Ruge-Kutte integration with a fixed step size of 0.1s had the same results as ODE45. Other integration methods may yield better results, but this remains unknown.

Added Mass and Inertia Matrices

It is recommended to include added mass and added inertia effects of the wings in addition to the hull in the total added mass and inertia matrices. The current model estimates the hull contribution using the prolate spheroid approximation, which is appropriate for capturing the dominant hydrodynamic behavior of the main body. However, this approach neglects the contribution of the wings.

To improve model fidelity, each wing should be modeled separately as an infinitely thin rectangular plate using Brennen's B9 table of added mass values (<http://brennen.caltech.edu/fluidbook/basicfluidynamics/unsteadyflows/addedmass/valuesoftheaddedmass.pdf>). This approximation is appropriate because the wings are thin appendages with finite span and chord, so they are better represented as plates rather than as part of the hull. Each wing's added mass should be estimated in the direction normal to the wing surface, transformed into the body fixed frame, and shifted to the vehicle center of gravity before being added to the full glider added mass and inertia matrices.

Including the wings would improve the model by capturing their effect on vertical motion, pitch, roll, and yaw/heading. It would also make the model more useful for fault cases, since partial or complete wing loss would change both the symmetry and hydrodynamic inertia of the glider. Thus, wing loss cases will have different added mass and inertia matrices than the nominal cases. This is not currently implemented in the dynamics inputs and will have to be implemented if wings are included in these matrices.

Artificial Damping

Hydrodynamic damping coefficients were not calculated. They may be useful to implement, whether found experimentally or calculated. One may choose to run some system identification and add them artificially, tuning as one sees fit.

Force and Moment Processing

Forces and moments are pulled from a lookup table of coefficients for each damage case, at the calculated AoA and sideslip angles. A more robust equation or set of equations for force and moment calculation and implementation may yield better results, at the cost of being significantly more work intensive.

I.C. Forces

The current forces modeler in the 2026 ROFASO Capstone Damage Assessment Protocol (D.A.P.) uses computational fluid dynamics and 3x3 wind tunnel testing to calculate what is needed to simulate glider physics/forces. This section will provide recommendations by the forces team to help improve the accuracy of the current model.

Computational Fluid Dynamics (CFD)

Computational Fluid Dynamics, or CFD, was used to help gather the coefficients of the hydrodynamic forces of lift and drag, as well as the coefficients of pitch, roll, and yaw moments. The program used was StarCCM+ (version 2506).

Some recommendations to improve the simulation and therefore improve the outputs' (coefficients') accuracy are as follows:

1. Create a high(er) fidelity mesh for the fluid volume region and Seaglider. High fidelity is a relative term and depends on the type of simulation being run. For the case of laminar flow around a Seaglider under constant density, a good place to start is a mesh of 2,000,000+ (two million) mesh cells.

- (a) The mesh cell count for the simulation used for the 2026 ROFASO Capstone project is approximately 222,000 cells, which is an order of magnitude off from the recommendation.
 - (b) With a large cell count, the usage of StarCCM+ locally through the university's VDI is difficult due to computational power. In order to properly run a simulation of this level fidelity (2 million+ cells), use of the HYAK Klone, the university's high powered computing cluster, is **strongly** recommended.
2. Perform the test cases with 3,000 (three thousand) or more iterations. This will allow the outputs/coefficients to converge to a single value.
- (a) The current iteration count for the 2026 ROFASO Capstone was 1,200 (twelve hundred). This amount was chosen due to time constraints. While a lot of the data converged within this amount of iterations, some – such as pitch, roll, and yaw moments – did not converge and oscillated between many values. This created values that did not have a specific trend when plotted against factors like angle of attack and side slip angle.
 - (b) The use of HYAK will allow automation for this many iterations as well as a quicker run time.

Physical Testing

The current and only physical testing for the forces and moments which act upon the Seaglider was use of the Aerodynamic 3x3 Wind Tunnel. These tests were performed under course angle of attack and side slip angle angle sweeps due to time constraints, as well as in a non-Reynolds number matches air flow.

Some recommendations for the physical testing are as follows:

- 1. Use of a water tunnel. This is like a wind tunnel except instead of air it is water. This should give a more accurate representation of the physics of the Seaglider in its actual operating environment.
 - (a) The use of the water tunnel was not used for the 2026 ROFASO Capstone.
 - (b) This utilization of the water tunnel can also help validate both CFD and 3x3 wind tunnel testing.
- 2. Completion of a second wind tunnel test. The wind tunnel test results from this test campaign are suspect. The following changes are recommended:
 - (a) Use of a larger, 3-D printed model with machined, metal wings and rudders. A smaller model was chosen such that the machined aluminum body would not exceed the limits

of the sting balance. This smaller model led to interactions with the balance mount, making strut tares less than useful, with negative drag at higher AoA and sideslip angle combinations. The 3-D printed wings and rudders had poor quality trailing edges. A larger, lighter model with machined wings would be able to mitigate these issues.

- (b) A sanity check hand calculation of skin friction and induced drag to compare against the measured drag. The flat-plate balance mount registers high drag and applying a strut tare

Ocean Current Modeling

In the current model of the Seaglider, the forces exerted on the glider by ocean currents have only been taken into account by trimming the first and last 20% of a dive profile. The Seaglider is modeled, as most basic gliders are modeled, as a suspended particle, with respect to the surrounding fluid.

The recommendation is obtaining a current model with the help of the Student Seaglider Center (SSC) and implementing it into the physics model of the D.A.P. system.

Researchers at Tsinghua University developed equations for current modelling in the paper "An optimization method for control parameters of underwater gliders considering energy consumption and motion accuracy"

II. Diagnoser

Dive Grouping Analysis: Group dives to produce a single representative result across the five-dive set. This includes making sure that the pilot inputs/actuators/parameters are the same. This is so the Diagnoser and Digital Twin can properly work together.

Diagnoser Tolerance Training: Train and tune tolerance values using a larger and more diverse dive set. This includes adding more dives outside of the Edmonds dives from April 2026, the MOTIVE mission dives, and the Colvos missions. If you know for sure that a dive is nominal this can also be used to tune the metrics so the Diagnoser can properly read what is considered nominal.